

Experiment No. 10

AIM: To study the characteristics of Zener diode.

APPARATUS: Zener diode kit, Ammeter, Voltmeter, Connecting wire.

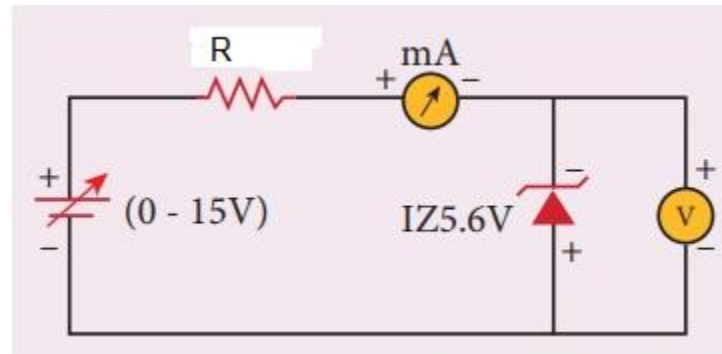
THEORY:

Zener Diode: Zener diode is a heavily doped PN junction diode. Due to heavily doped, its depletion layer is very thin and is order of micrometer. The forward bias characteristic of Zener diode is same as the normal PN junction diode but in reverse bias it has different characteristic. Initially, a negligible constant current flow through the zener diode in its reverse bias but at certain voltage, the current becomes abruptly large. This voltage is called as zener voltage. This sudden and sharp increase in zener current is called as zener breakdown.

PROCEDURE:

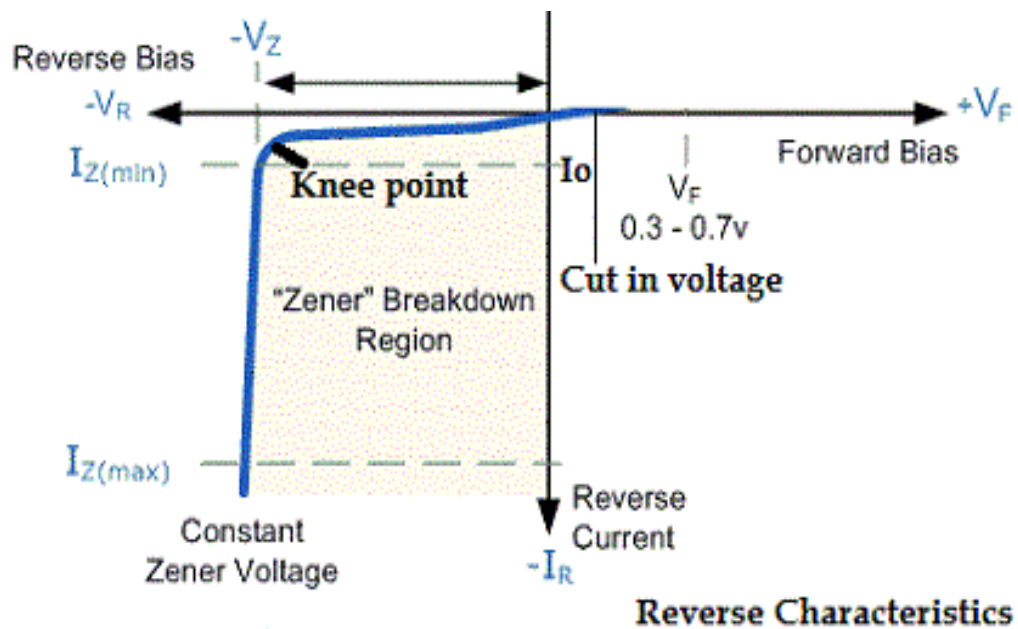
- 1) In the reverse bias, the polarity of the DC power supply is reversed so that the P- region of the diode is connected to the negative terminal and N-region to the positive terminal of the DC power supply.
- 2) The connections are made as given in the circuit diagram.
- 3) The voltage across the diode can be varied with the help of the variable DC power supply.
- 4) The reverse voltage (V_R) across the diode is increased from 0.5V in steps of 0.5V up to 6V and the reverse current (I_R) through the diode is noted from the milli-ammeter. The readings are tabulated.
- 5) Initially, the voltage is increased in steps of 0.5V. When the breakdown region is approximately reached, then the input voltage may be raised in steps of, say 0.1V to find the break-down voltage.
- 6) The reverse voltage and reverse current are taken as negative.
- 7) A graph is drawn taking the reverse bias voltage along negative x-axis and the reverse bias current along negative y-axis.
- 8) In the reverse bias, Zener breakdown occurs at a particular voltage called Zener voltage V_Z (~5.6 to 5.8V) and a large amount of current flows through the diode which is the characteristics of a Zener diode.
- 9) The breakdown voltage of the Zener diode is determined from the graph as shown.

DIAGRAM:



OBSERVATION TABLE
REVERSE BIASED:

S.NO	REVERSE VOLTAGE (V)	REVERSE CUREENT(ma)
1.		
2.		
3.		
4.		
5		
6		
7		
8		
9		
10		



RESULT: The V-I characteristics of the Zener diode are studied, The breakdown voltage of the Zener diode $V_z = \text{----V}$

PRECAUTIONS:

1. While doing the experiment do not exceed the ratings of the diode. This may lead to damage of the diode.
2. Connect voltmeter and Ammeter in correct polarities as shown in the circuit diagram.
3. Do not switch ON the power supply unless you have checked the circuit connections as per the circuit diagram.

VIVA VOCE:

1. What is the need for doping?
2. How depletion region is formed in the PN junction?
3. What is leakage current?
4. What is break down voltage?
5. What is an ideal diode? How does it differ from a real diode?
6. What is the effect of temperature in the diode reverse characteristics?
7. What is cut-in or knee voltage? Specify its value in case of Ge or Si?
8. What are the difference between Ge and Si diode?
9. What is the capacitance formed at forward biasing?
10. What is the relationship between depletion width and the concentration of impurities?